

PRODUCT DEMO DOCUMENT

Denial Doctor Platform

AI-powered denial analysis and correction platform for healthcare revenue cycle management. Instant health scanning, multi-agent correction pipeline, and real-time claim resubmission delivering 4-5x higher recovery rates at a fraction of traditional costs.

Prepared by Denial Doctor Product Team

May 2026

Version 1.0

CONFIDENTIAL

Table of Contents

1. Executive Summary	2
2. How Denial Doctor Works	3
2.1 Phase 1: Instant Denial Health Scan (Pre-Contract)	3
2.2 Phase 2: Active Claim Correction and Resubmission (Post-Contract)	4
3. Why Denial Doctor Is Superior	6
3.1 AI-First Architecture vs. Rule-Based Systems	6
3.2 Two-Phase Engagement Model: Prove Value Before Commitment	7
3.3 Multi-Agent Specialization vs. Monolithic Processing	7
3.4 Real-Time Processing vs. Batch-Mode Operations	8
3.5 Dramatic Cost Efficiency Gains	8
4. Technology Architecture	9
4.1 AI Model Stack	9
4.2 Security and Compliance	9
5. Product Roadmap and Future Expansion	10
6. Conclusion	11

1. Executive Summary

Denial Doctor is an AI-powered denial management platform purpose-built for healthcare revenue cycle management (RCM). In the United States alone, healthcare payers deny approximately \$262 billion in claims annually, and providers recover less than 15% of that lost revenue through traditional manual appeals processes. Denial Doctor transforms this equation by deploying advanced artificial intelligence to instantly analyze denial patterns, generate actionable corrective strategies, and accelerate claim resubmission with precision that manual workflows simply cannot match. The platform is designed to serve billing companies, RCM firms, hospitals, and physician practices that are drowning in denied claims and struggling with aging accounts receivable.

At its core, Denial Doctor addresses a critical industry gap: the latency between denial receipt and corrective action. Traditional workflows require experienced coders and billing specialists to manually review each denied claim, identify the root cause, research payer-specific requirements, and prepare a corrected claim or appeal letter. This process typically takes 30-45 minutes per claim and suffers from inconsistency, high labor costs, and error rates that compound across thousands of denials. Denial Doctor reduces this cycle from hours to seconds by automating the entire analysis-to-correction pipeline while maintaining compliance with payer-specific rules and regulatory requirements.

The platform operates in two distinct engagement phases. In the first phase, prospective clients upload their denial data and receive an instant health assessment with a quantitative rating, pinpointing exactly where revenue leakage occurs. This pre-contract scan is designed to build trust and demonstrate value before any financial commitment. In the second phase, once a contract is signed, Denial Doctor activates its full suite of AI agents to systematically correct and resubmit denied claims, recovering revenue that would otherwise be written off as bad debt.

Metric	Industry Average	With Denial Doctor
Denial-to-Correction Time	30-45 min/claim	< 60 sec/claim
Denial Recovery Rate	12-15%	65-80%
Cost per Corrected Claim	\$25-40	\$3-7
First-Pass Resubmission Acceptance	55-60%	85-92%
Staff Hours per 1,000 Denials	500-750 hours	40-80 hours

Table 1: Denial Doctor Performance Benchmarks vs. Industry Average

2. How Denial Doctor Works

Denial Doctor operates through a streamlined, two-phase engagement model that mirrors the real-world sales cycle of RCM service providers. The platform was architected with the understanding that selling denial management services requires demonstrating tangible value before a client commits. Each phase is designed to be self-contained, yet they flow seamlessly into one another, creating a natural progression from discovery to resolution. The entire system is powered by Azure-hosted AI models including GPT-5.5 and GPT-5.3-Codex, ensuring enterprise-grade reliability and HIPAA-compliant infrastructure.

2.1 Phase 1: Instant Denial Health Scan (Pre-Contract)

The first phase is the client-facing discovery tool. When a prospective client uploads their denial report (typically an 835 remittance CSV export from their practice management system), Denial Doctor performs a comprehensive AI-driven analysis in real time. This scan is designed to accomplish two objectives: first, to provide the client with an immediate, quantitative understanding of their denial landscape; and second, to demonstrate the sophistication and accuracy of the platform, creating a compelling business case for engagement.

Step-by-Step Process:

- â■ CSV Upload and Ingestion: The client drags and drops their denial CSV file into the Denial Doctor portal. The system accepts standard 835/ERA formats, custom practice management exports, and flat-file denial reports. The upload interface supports files up to 500 MB and automatically detects column mappings, date formats, and payer identifiers. No manual formatting or data preparation is required from the client.
- â■ Automated Data Parsing and Enrichment: Upon upload, the AI engine parses every row, mapping denial reason codes (CARC/RARC) to their standardized descriptions, cross-referencing payer-specific policies, and enriching each record with contextual data such as claim age, dollar amount at risk, and historical denial frequency for the same reason code. The system identifies duplicates, flags incomplete records, and normalizes data inconsistencies across multiple payer formats.
- â■ Denial Pattern Recognition: Using advanced clustering algorithms, the AI identifies recurring denial patterns across the dataset. Rather than treating each denial in isolation, Denial Doctor groups related denials by root cause, enabling the platform to identify systemic issues such as a specific payer consistently rejecting a particular procedure code, or a recurring eligibility verification gap that affects an entire provider group.
- â■ Health Score Generation (x/10 Rating): The platform computes a composite Denial Health Score on a scale of 1 to 10, where 10 represents an exceptionally clean claims submission process with minimal denials, and 1 indicates severe systemic issues requiring immediate intervention. The score is derived from multiple weighted factors including overall denial rate, denial aging distribution, root cause concentration (whether denials are spread across many causes or

concentrated in a few fixable areas), and revenue-at-risk distribution. A score of 7 or above typically indicates a well-managed revenue cycle with room for targeted improvements, while a score below 5 signals critical operational gaps that are likely costing the organization significant revenue.

âââ **Key Issue Identification:** Alongside the score, the platform generates a prioritized list of the top denial categories, each accompanied by the number of affected claims, total dollar value at risk, and an estimated recovery potential. Issues are ranked by a composite of financial impact and correction feasibility, so clients see not just what is broken, but what is most worth fixing first. This prioritization is critical because it transforms a raw data dump into a strategic roadmap for revenue recovery.

Denial Category	Claims	Amount at Risk	Recovery Potential	Priority
Eligibility / Coverage Verification	847	\$1,284,500	78%	HIGH
Authorization / Prior Auth Missing	523	\$892,100	72%	HIGH
Coding Errors (ICD/CPT Mismatch)	412	\$623,400	85%	MEDIUM
Duplicate Claim Submissions	198	\$287,600	91%	LOW
Timely Filing Violations	156	\$412,300	35%	MEDIUM

Table 2: Sample Instant Denial Health Scan Results

The Instant Denial Health Scan report is formatted as a client-ready deliverable, suitable for presentation in sales meetings, board reviews, or executive briefings. The report includes the overall health score, denial distribution charts, top issue breakdown, and a financial impact summary. This document serves as both a diagnostic tool and a persuasive sales asset, clearly demonstrating the value of engaging Denial Doctor for full denial recovery services.

2.2 Phase 2: Active Claim Correction and Resubmission (Post-Contract)

Once a contract is executed, the client gains access to the full Denial Doctor correction engine. This is where the platform transitions from diagnostic mode to active remediation, systematically working through the denied claims identified in the health scan and preparing corrected claims for resubmission. The correction engine operates through a multi-agent architecture where each AI agent specializes in a specific aspect of the denial resolution workflow, ensuring depth of expertise and accuracy in each domain.

The Five-Agent Correction Pipeline:

Agent	Specialization	Key Functions
1. Eligibility Agent	Eligibility, Demographics, and Prior Authorization	Verifies patient eligibility at time of service, corrects demographic errors, validates or obtains prior authorization numbers, and cross-references payer enrollment databases
2. Coding Agent	Medical Coding and Supporting Documentation	Corrects ICD-10/CPT/HCPCS code mismatches, ensures medical necessity linkage, identifies missing modifiers, and generates supporting documentation templates
3. Scrubbing Agent	Claim Scrubbing and Payer ID Validation	Validates payer IDs, runs pre-submission claim scrubbing against payer-specific edit rules, ensures correct bill type and frequency codes, and verifies NPI/Taxonomy combinations
4. Payment Agent	Payment Posting and Secondary Insurance	Posts payments and adjustments, identifies zero-pay denials, calculates patient responsibility vs. payer responsibility, and prepares secondary/tertiary insurance claims
5. Denial Agent	Denial Analysis and Appeal Generation	Root cause analysis for each denial, generates payer-specific appeal letters with supporting evidence, tracks timely filing deadlines, and manages appeal escalation workflows

Table 3: The Five-Agent Correction Pipeline

Each agent operates semi-autonomously but is coordinated by the Denial Agent, which serves as the central orchestrator. When a denied claim enters the correction pipeline, the Denial Agent first classifies the root cause and routes it to the appropriate specialist agent. That agent performs its domain-specific analysis and correction, then returns the corrected claim data to the Denial Agent for final review. If the correction requires cross-domain expertise (for example, a claim denied for both coding errors and missing authorization), the Denial Agent coordinates multiple agents in sequence before assembling the final corrected claim.

The correction workflow includes multiple quality checkpoints. Before any claim is resubmitted, a validation layer verifies that all corrective actions are consistent with the original denial reason, that no new errors have been introduced during correction, and that the resubmission complies with the specific payer's filing requirements and timelines. This multi-layered quality assurance ensures that corrected claims have the highest possible probability of acceptance on first resubmission, minimizing the costly cycle of rework that plagues manual correction processes.

3. Why Denial Doctor Is Superior

The denial management market is crowded with legacy software vendors and outsourcing companies that offer partial solutions. Denial Doctor differentiates itself through a combination of architectural innovation, AI-first design philosophy, and a business model that aligns platform incentives with client outcomes. The following analysis examines the specific competitive advantages that make Denial Doctor the superior choice for healthcare organizations seeking to maximize denial recovery rates while reducing operational costs.

3.1 AI-First Architecture vs. Rule-Based Systems

Legacy denial management tools rely on static rule engines and pre-defined workflows that require constant manual updates as payer policies change. These rule-based systems are inherently reactive: they can only handle denial scenarios that have been explicitly programmed, and they break silently when encountering new denial patterns. In contrast, Denial Doctor is built on a foundation of large language models (GPT-5.5 and GPT-5.3-Codex) that can reason about denial causes, interpret nuanced payer-specific policies, and generate contextually appropriate corrections without requiring explicit programming for each scenario.

This architectural difference has profound practical implications. When a payer introduces a new denial reason code or changes its authorization requirements, rule-based systems require manual rule creation, testing, and deployment, a process that typically takes 2-4 weeks. During that lag, every claim hitting the new rule is either mishandled or ignored. Denial Doctor's AI models adapt to new patterns within the data itself, recognizing novel denial scenarios and generating corrections based on contextual understanding of payer behavior, medical coding standards, and regulatory requirements. This translates to near-zero lag time between payer policy changes and effective denial correction.

Capability	Rule-Based Systems	Denial Doctor AI
New denial pattern recognition	Requires manual rule creation (2-4 week lag)	Automatic pattern detection and correction
Payer policy changes	Breaks until rules updated	Adapts through contextual reasoning
Complex multi-code denials	Often fails on edge cases	Handles cross-domain reasoning
Appeal letter generation	Template-based, generic	Contextual, payer-specific, evidence-cited
Scalability	Degrades with volume	Improves with more data
Maintenance cost	\$50K-150K/year for rule updates	Minimal; models self-improve

Table 4: AI-First Architecture vs. Rule-Based Systems

3.2 Two-Phase Engagement Model: Prove Value Before Commitment

Most denial management vendors require clients to commit to a contract before demonstrating any measurable value. This creates a significant barrier to adoption, especially for mid-market billing companies and physician practices that are understandably skeptical of vendor claims. Denial Doctor eliminates this friction through its two-phase engagement model, which provides tangible, quantifiable value before any financial commitment is required.

In Phase 1, the Instant Denial Health Scan gives clients a comprehensive, visually compelling report showing exactly where their revenue is leaking and how much is recoverable. This is not a superficial demo with sample data; it is a real analysis of the client's actual denial portfolio. The health score provides an immediate, objective benchmark, and the prioritized issue list gives the client an actionable roadmap regardless of whether they choose to engage Denial Doctor for correction services. This transparency builds trust and dramatically shortens the sales cycle, because clients can see precisely what they stand to gain before signing a contract.

Competitive vendors that offer free assessments typically provide only high-level summaries with limited actionability, designed primarily as lead-generation tools. Denial Doctor's Phase 1 scan delivers diagnostic depth that rivals what most vendors charge for as a full engagement. This is only possible because the AI analysis pipeline is fully automated and can process thousands of claims in minutes, a capability that manual assessment workflows cannot match at any price point.

3.3 Multi-Agent Specialization vs. Monolithic Processing

A critical design decision in Denial Doctor is its multi-agent architecture, where separate AI agents specialize in distinct domains of the denial correction workflow. This stands in stark contrast to monolithic AI approaches that attempt to handle all denial types with a single model or workflow. The multi-agent design delivers superior accuracy because each agent can develop deep domain expertise in its specialization, much like how human specialists outperform generalists in complex tasks.

Consider a claim denied for both an authorization mismatch and a coding error. A monolithic system might address the coding error while missing the authorization issue, or vice versa, because it lacks the specialized reasoning paths for each domain. Denial Doctor's Eligibility Agent focuses exclusively on authorization and eligibility verification, while the Coding Agent specializes in ICD/CPT code relationships and medical necessity logic. The Denial Agent coordinates both, ensuring that all identified issues are resolved before the corrected claim is assembled. This approach yields significantly higher first-pass resubmission acceptance rates because corrections are more thorough and more accurate.

The multi-agent architecture also provides natural extensibility. As new denial categories emerge or as the platform expands into adjacent RCM domains (such as charge capture optimization or underpayment detection), new specialist agents can be added without disrupting existing workflows. This modularity ensures that Denial Doctor can evolve with the healthcare reimbursement landscape

without requiring fundamental architectural changes, protecting the client's investment over the long term.

3.4 Real-Time Processing vs. Batch-Mode Operations

Many denial management platforms operate on batch processing schedules, where claims are queued and processed in nightly or weekly runs. This introduces latency that has both operational and financial consequences. From an operational standpoint, batch processing means that denial correction status is always at least one cycle behind reality, making it difficult for managers to get an accurate picture of current denial inventory. From a financial standpoint, every day of delay in correcting and resubmitting a denied claim is a day that the organization is carrying that amount as accounts receivable, increasing the risk of write-offs and reducing cash flow velocity.

Denial Doctor processes denials in real time. As soon as a denial report is uploaded, the AI engine begins analysis immediately, and corrected claims are generated within seconds of processing. This real-time capability is particularly valuable for claims approaching timely filing deadlines, where even a single day of processing delay can mean the difference between a successful appeal and a permanent revenue loss. The platform includes proactive deadline tracking that automatically prioritizes claims at risk of timely filing expiration, ensuring that the most time-sensitive denials are addressed first.

3.5 Dramatic Cost Efficiency Gains

The economics of manual denial management are straightforward but unsustainable. At an average cost of \$25-40 per corrected claim (including staff time, overhead, and management supervision), a mid-size billing company processing 10,000 denials per month spends \$250,000-400,000 monthly on denial correction alone. Denial Doctor reduces this cost to \$3-7 per corrected claim, representing a 75-85% reduction in per-claim correction costs. For the same 10,000-denial volume, this translates to a monthly cost of \$30,000-70,000, saving between \$180,000 and \$370,000 per month.

Beyond direct labor savings, Denial Doctor generates significant indirect cost benefits. Faster denial correction means shorter accounts receivable cycles, which improves cash flow and reduces the need for working capital financing. Higher first-pass resubmission acceptance rates eliminate the compounding costs of rework, where each failed resubmission adds another 15-30 days to the revenue cycle. And because the AI operates consistently across all claims, the variance in correction quality that plagues manual workflows (where experienced staff produce better results than new hires) is eliminated, ensuring uniform quality regardless of claim volume or staff turnover.

Cost Factor	Manual Process	Denial Doctor	Savings
Per-claim correction cost	\$25-40	\$3-7	75-85%
Monthly cost (10K denials)	\$250K-400K	\$30K-70K	\$180K-370K
Average AR days for denials	45-65 days	12-20 days	60-70%

Staff required per 10K denials	12-18 FTEs	2-4 FTEs	78-83%
Annual training/onboarding cost	\$45K-80K	\$5K-10K	85-88%

Table 5: Cost Comparison - Manual Process vs. Denial Doctor

4. Technology Architecture

Denial Doctor is built on Microsoft Azure's enterprise cloud infrastructure, leveraging Azure OpenAI Service for AI model hosting, Azure Active Directory for identity management, and Azure Virtual Machines for application deployment. This architecture provides the security, scalability, and compliance foundation required for processing protected health information (PHI) in a HIPAA-regulated environment.

4.1 AI Model Stack

The platform utilizes a tiered AI model strategy where different models are deployed for different tasks based on their strengths. GPT-5.5 serves as the primary reasoning engine for denial analysis, root cause classification, and corrective action generation. GPT-5.3-Codex handles structured data transformation tasks such as CSV parsing, claim data normalization, and correction output formatting. This model specialization ensures that each task is handled by the model best suited for it, maximizing both accuracy and processing speed. All model interactions occur over encrypted connections within Azure's private network, ensuring that PHI never traverses public internet pathways.

Model	Role	Tasks
GPT-5.5	Primary Reasoning Engine	Denial root cause analysis, correction strategy generation, appeal letter drafting, cross-domain reasoning
GPT-5.3-Codex	Structured Data Engine	CSV parsing, claim normalization, code validation, output formatting, data transformation pipelines
DeepSeek-V4-Flash	Validation Engine	Pre-submission claim scrubbing, compliance verification, quality assurance checks

Table 6: AI Model Stack and Role Assignments

4.2 Security and Compliance

All data processing occurs within Azure's HIPAA-compliant infrastructure. Client denial data is encrypted at rest using AES-256 and in transit using TLS 1.3. The platform implements role-based access control (RBAC) through Azure Active Directory, ensuring that each user can only access data and functions appropriate to their role. Audit logs capture all data access and modification events, providing a complete chain of custody for compliance reporting. The multi-tenant architecture provides logical data isolation between clients, ensuring that no client can access another client's data under any circumstances. Regular penetration testing and security assessments validate the platform's security posture against evolving threats.

5. Product Roadmap and Future Expansion

Denial Doctor Phase 1 focuses exclusively on denial management, delivering a best-in-class solution for this critical RCM function. However, the platform's multi-agent architecture is designed for progressive expansion across the entire revenue cycle. Each additional agent module builds on the shared data layer, AI infrastructure, and client portal, creating a comprehensive RCM intelligence platform over time. The roadmap reflects a deliberate strategy of mastering one domain before expanding to adjacent ones, ensuring that each new module meets the same quality and accuracy standards that define the denial management product.

Phase	Agent Module	Capability	Timeline
1	Denial Management	Denial analysis, correction, and resubmission with instant health scanning	Available Now
2	Eligibility Verification	Real-time eligibility checks, demographic validation, and prior authorization management	Q3 2026
3	Coding Optimization	AI-assisted medical coding with charge capture optimization and documentation improvement	Q4 2026
4	Claim Scrubbing	Pre-submission claim validation with payer-specific edit rules and compliance checks	Q1 2027
5	Payment Posting	Automated payment posting, underpayment detection, and secondary insurance coordination	Q2 2027
6	Quality Assurance	Cross-agent quality monitoring, performance benchmarking, and continuous improvement feedback loops	Q3 2027

Table 7: Denial Doctor Product Roadmap

The Quality Assurance agent in Phase 6 serves as the capstone of the platform, continuously monitoring the performance of all other agents, identifying patterns in correction failures, and feeding improvement signals back into the AI models. This creates a self-improving system where correction accuracy increases over time as the platform processes more claims and encounters more denial scenarios. The ultimate vision is a fully autonomous RCM intelligence platform that not only corrects errors after they occur but proactively prevents denials before claims are submitted, shifting the entire revenue cycle from reactive to predictive.

6. Conclusion

Denial Doctor represents a paradigm shift in healthcare denial management. By combining AI-first architecture with a client-friendly engagement model, the platform removes the traditional barriers to adoption while delivering performance that manual and rule-based systems cannot approach. The Instant Denial Health Scan provides immediate, tangible value at zero risk, while the multi-agent correction engine delivers recovery rates that are 4-5x higher than industry averages at a fraction of the cost.

In a healthcare environment where margins are increasingly compressed and regulatory complexity is growing, the ability to recover denied revenue efficiently is not a luxury but a necessity. Organizations that continue to rely on manual denial workflows or legacy rule-based tools will find themselves at an increasing competitive disadvantage as AI-native solutions like Denial Doctor redefine what is possible in revenue cycle management. The question is not whether AI will transform denial management; it is whether your organization will be an early adopter that captures the advantage or a late follower that must catch up.

To experience the Denial Doctor Instant Denial Health Scan with your own denial data, contact our team to schedule a demonstration. There is no cost, no commitment, and no risk. Upload your denial report and see your health score in minutes.